

User Guide: Performance Reporting Dashboard

1. Introduction

The Performance Reporting R Shiny Dashboard is designed for comprehensive financial analysis, allowing you to assess the performance of traditional funds, hedge funds, and custom-blended portfolios. You can conduct detailed risk analysis, including advanced simulations, and evaluate performance metrics like the Sharpe Ratio.

2. Setup and Launch Instructions

To ensure the application runs correctly, please follow these setup steps carefully.

Step 1: File and Folder Structure

All application files should be placed in the same directory (folder) as your RStudio project. This includes:

- **global.R**
- **ui.R**
- **server.R**
- **app.R**
- **All required data files** (e.g., .csv or .rds files containing your fund and index data).

Step 2: Install Required Libraries

Before the first launch, you need to install the necessary R packages. Open the **global.R** file in **RStudio**. It contains a list of all required libraries (like shiny, shinydashboard, xts, ggplot2, etc.). Install any missing packages by running the **install.packages** command in the RStudio console for each one.

Step 3: Running the Application

The application must be launched in a specific order:

1. Run **global.R** first: This loads all the necessary data and libraries into your R session.
2. Run **ui.R**: This loads the user interface structure.
3. Run **server.R**: This loads the application's backend logic.
4. Run **app.R**: This final step launches the application, which will open in the RStudio Viewer pane or a new browser window.

3. How to Use the Dashboard

The dashboard is organized into several tabs, which you can select from the sidebar menu on the left.

A. Traditional & Hedge Fund Performance

- On these tabs, use the "Select Fund" dropdown menu to choose the specific traditional or hedge fund you wish to analyze.
- Use the "Select Date Range" calendar to focus your analysis on a specific time period. The charts and tables will update automatically based on your selections.

B. Portfolio Blending

This tab allows you to create and analyze a custom two-fund portfolio.

- **Fund Selection & Weights:** Select two different funds from the dropdown menus and assign a weight to each using the numeric input fields (e.g., 0.60 for 60%).
- **Weight Sum Check:** A critical feature on this tab is the "Weight Sum Check" box. If the weights you assign to the two funds do not add up to exactly 1.00 (or 100%), a warning message will appear to alert you. This ensures your portfolio allocation is valid.

C. Risk Analysis.

- **Select Portfolio Type:** First, choose whether you want to analyze a single "Traditional Fund" or the "Blended Portfolio" you configured in the previous tab. The available inputs will change based on your choice.
- **Set Analysis Parameters:** Define your analysis by setting the Horizon (in years) and the Number of Simulations. Higher simulation counts lead to more accurate results but may take longer to process.

D. Sharpe Ratio Analysis

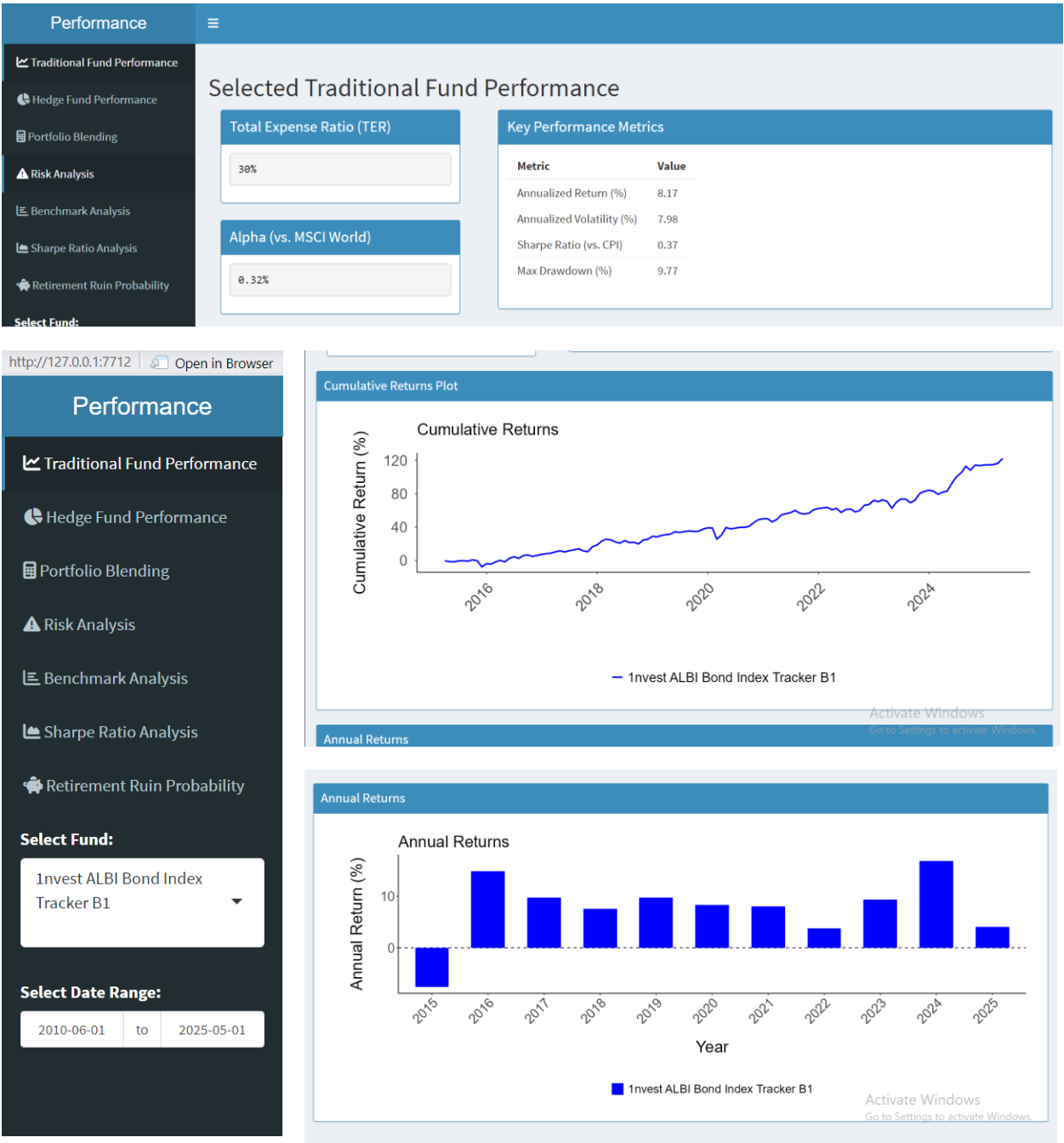
- Similar to the Risk Analysis tab, first select whether you want to analyze a "Traditional Fund" or a "Blended Portfolio".
- Set the "Rolling Window (Years)" to define the time period for the rolling Sharpe Ratio calculation. The plots will update accordingly.

E. Ruin Probability Simulation,

To run the **Ruin Probability Simulation**, you need to set all the required parameters, including selecting the portfolio type (either Traditional Fund or Blended Portfolio), setting the horizon (years), and specifying the number of simulations. After configuring these settings, click on the "**Run Simulation**" button. The application will then perform the simulation and display the **Cumulative Probability of Ruin** plot.

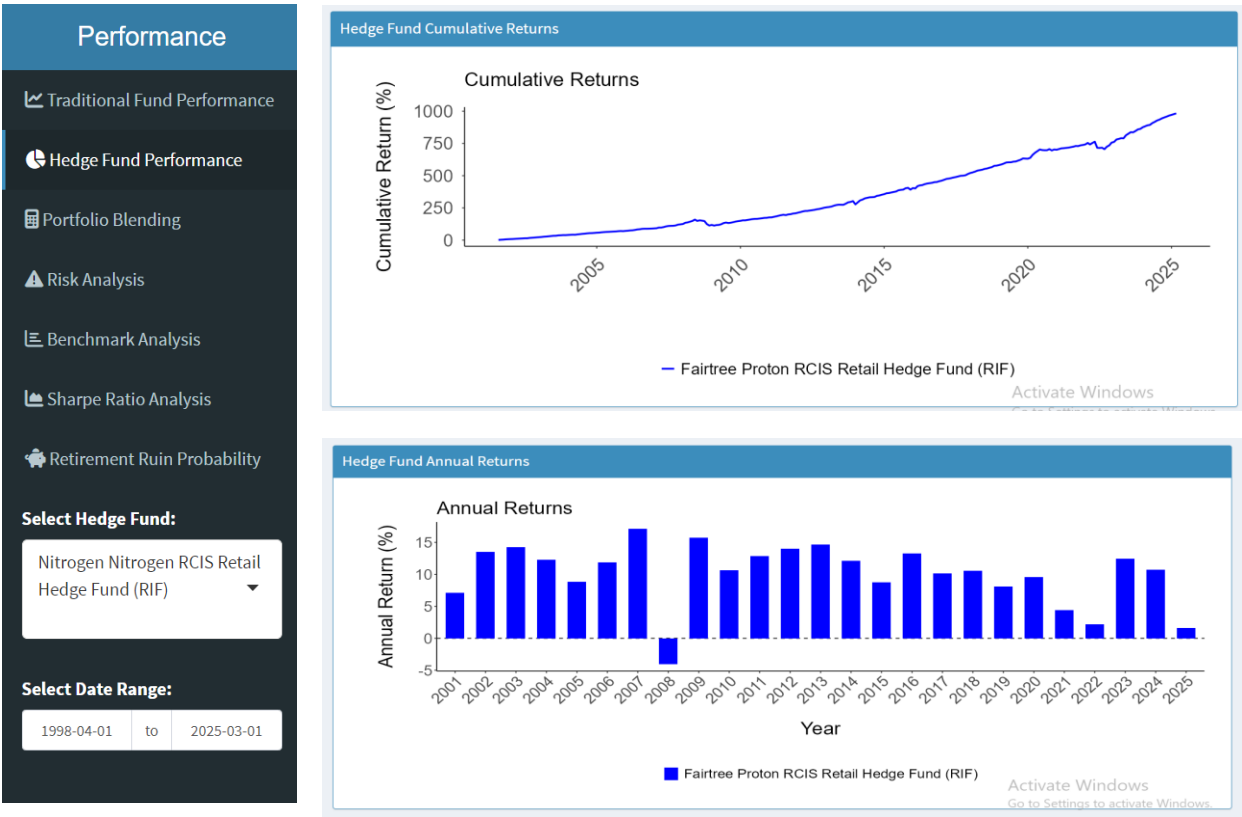
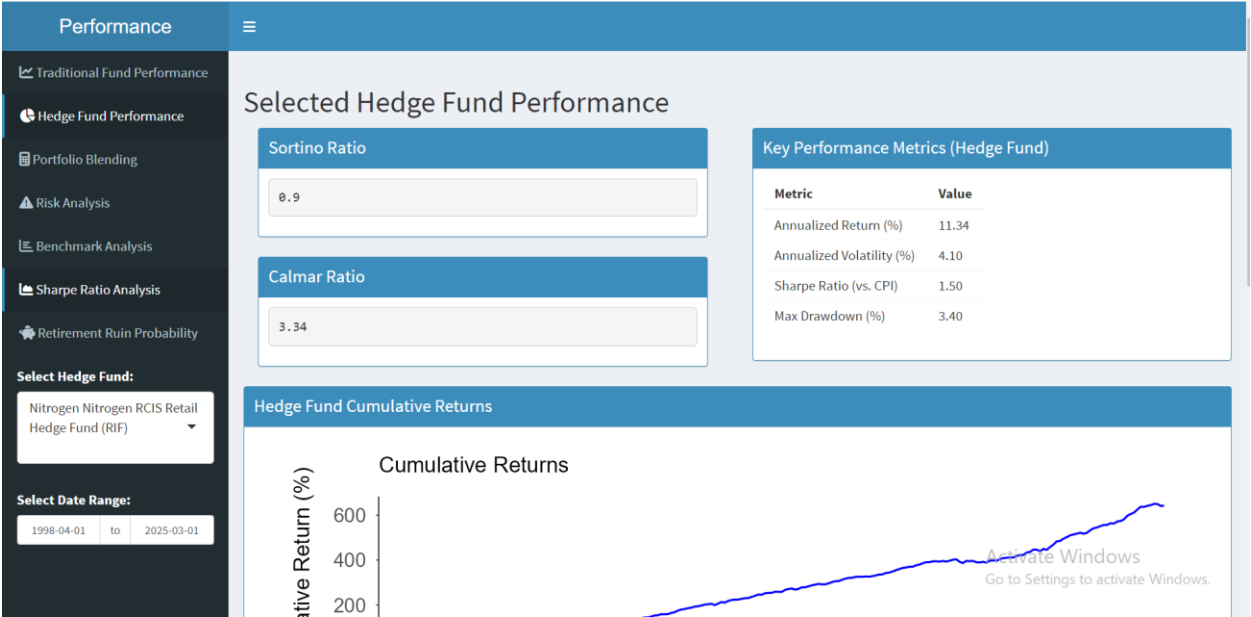
Traditional Fund Performance Tab

This tab provides a detailed performance overview for selected traditional funds. It displays key metrics such as Total Expense Ratio (TER) and Alpha (vs. MSCI World), alongside comprehensive cumulative and annual returns plots. This allows for a quick assessment of a fund's historical performance and cost efficiency.



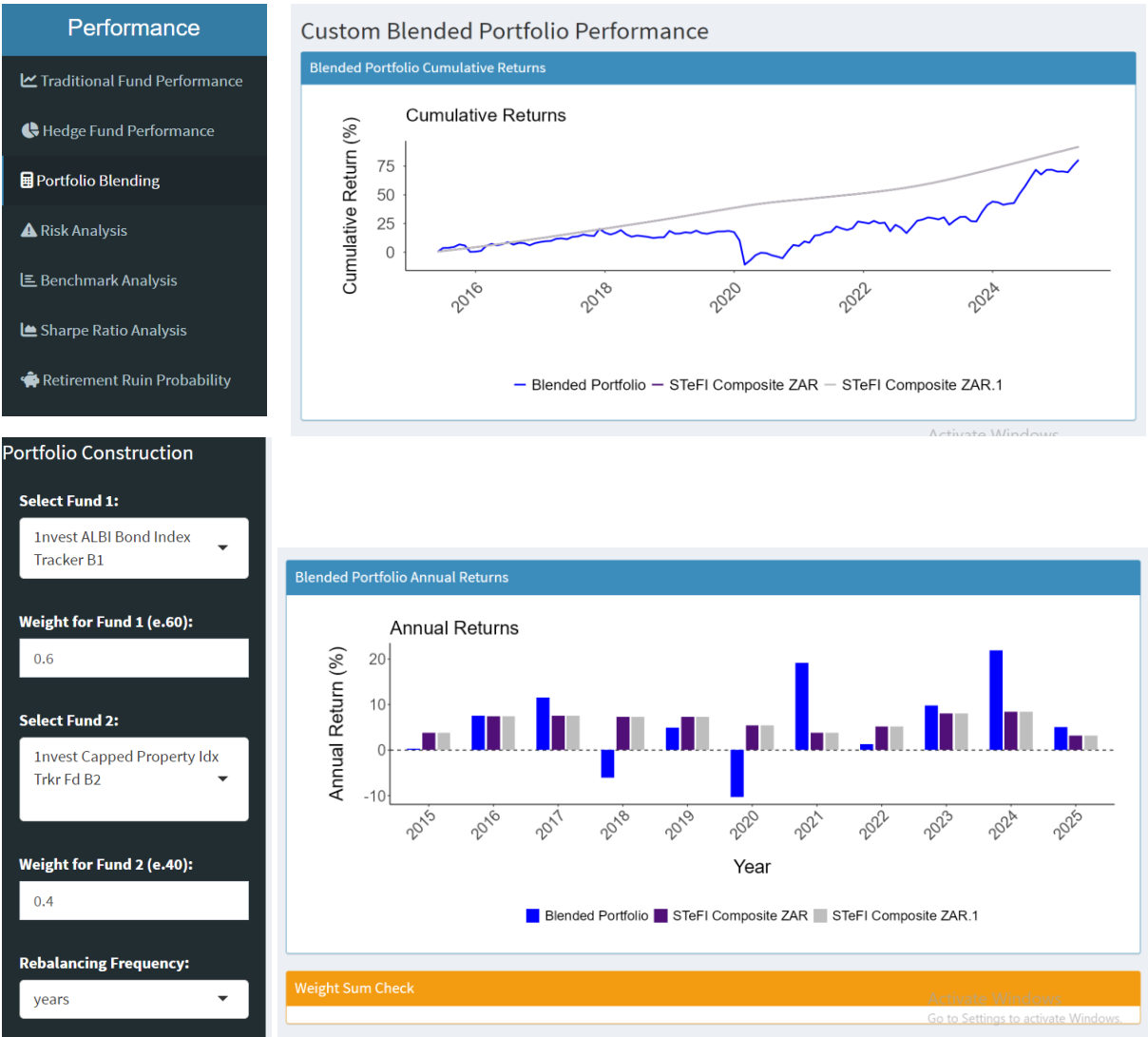
Hedge Fund Performance

Tab hedge fund analysis, this tab offers insights into their risk-adjusted returns. It features metrics like Sortino Ratio and Calmar Ratio, along with a summary of key performance indicators. Cumulative and annual returns plots are also provided for a complete performance picture.



Portfolio Blending

This tab enables users to construct and analyze custom blended portfolios by combining two selected funds with specified weights and rebalancing frequencies. It visualizes the blended portfolio's cumulative and annual returns, allowing for comparison against chosen benchmarks. A weight sum check ensures valid portfolio construction.



Risk Analysis

The Risk Analysis tab focuses on assessing the potential downside risk of selected portfolios (traditional or blended) against benchmarks. It includes plots for Max Drawdown Probability Density, Historical Drawdowns, and Rolling Returns over various horizons, providing a comprehensive view of risk characteristics.

Risk Analysis Settings

Select Portfolio Type:

Traditional Fund

Select Traditional Fund:

Invest ALBI Bond Index Tracker B1

Select Benchmarks & CPI for Comparison

Select ASISA Benchmark:

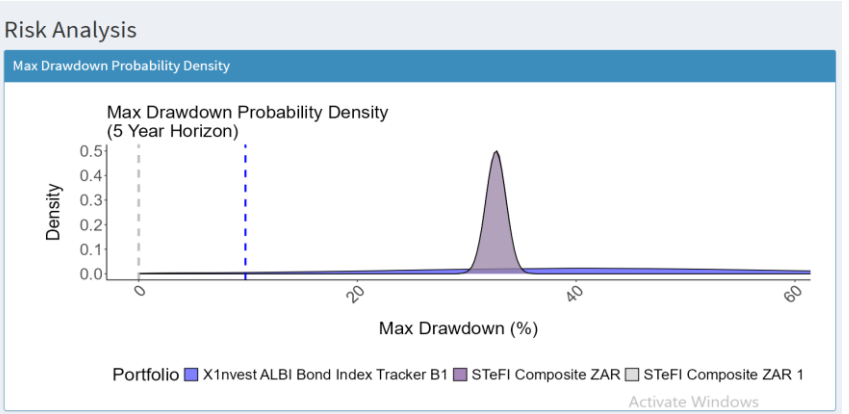
STeFI Composite ZAR

Select CPI Target:

STeFI Composite ZAR

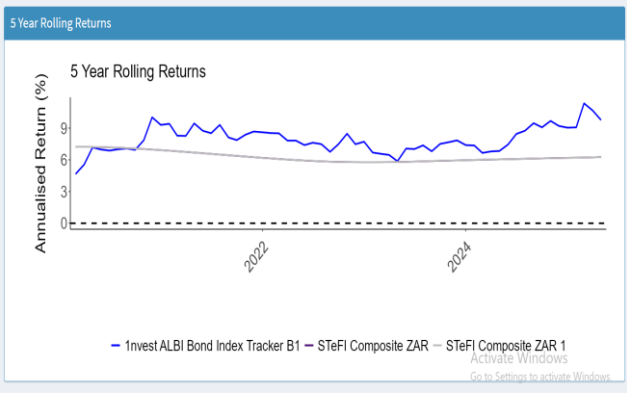
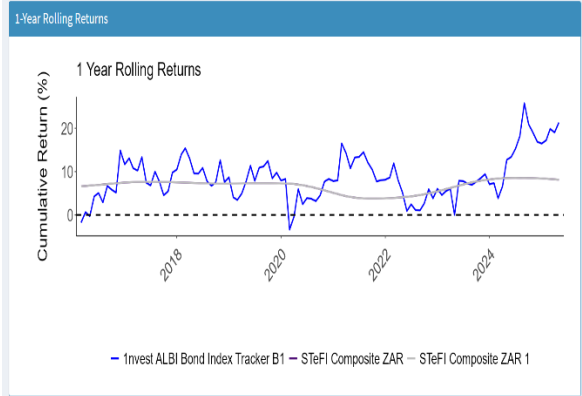
Horizon (Years):

5



Max Drawdown Probability Table

Portfolio	maxDD_range	frequency	median	probability
XInvest ALBI Bond Index Tracker B1	0 - 5	1794.00	2.73	1.79
XInvest ALBI Bond Index Tracker B1	5 - 10	2227.00	7.74	2.23
XInvest ALBI Bond Index Tracker B1	10 - 15	3252.00	12.77	3.25
XInvest ALBI Bond Index Tracker B1	15 - 20	4803.00	17.72	4.80
XInvest ALBI Bond Index Tracker B1	20 - 25	6520.00	22.72	6.52
XInvest ALBI Bond Index Tracker B1	25 - 30	8460.00	27.62	8.46
XInvest ALBI Bond Index Tracker B1	30 - 35	9925.00	32.61	9.93
XInvest ALBI Bond Index Tracker B1	35 - 40	10747.00	37.59	10.75
XInvest ALBI Bond Index Tracker B1	40 - 45	11122.00	42.43	11.13
XInvest ALBI Bond Index Tracker B1	45 - 50	10415.00	47.43	10.42
XInvest ALBI Bond Index Tracker B1	50 - 55	9074.00	52.35	9.08
XInvest ALBI Bond Index Tracker B1	55 - 60	7249.00	57.33	7.25
XInvest ALBI Bond Index Tracker B1	60 - 65	5408.00	62.27	5.41



Benchmark Analysis

This tab facilitates in-depth comparison of a selected portfolio (traditional or blended) against multiple benchmarks. It presents cumulative, annual, and rolling performance plots, along with a summary table of annualized returns. It also calculates and displays tracking error and information ratio to quantify relative performance.



Performance Summary vs. Benchmarks

	Annualized Return	Annualized Std Dev	Annualized Sharpe (Rf=0%)
Invest ALBI Bond Index Tracker B1	0.08	0.08	1.02
MSCI World NR USD	0.14	0.16	0.92

Sharpe Ratio Analysis

This tab provides a detailed examination of the Sharpe Ratio for a selected fund or blended portfolio. It presents both expanding and rolling window Sharpe Ratios, complete with confidence intervals, to illustrate the consistency and statistical significance of risk-adjusted returns over time.



Retirement Ruin Probability

Designed for long-term financial planning, this tab simulates the probability of portfolio "ruin" (depletion) under various withdrawal rates and inflation adjustments. Users can input an initial principal, time horizon, and number of simulations to assess the sustainability of their retirement withdrawal strategy

Retirement Simulation Settings

Select Fund:

Invest ALBI Bond Index Tracker B1

Initial Principal:

100000

Time Horizon (Years):

30

Number of Simulations:

100

Annual Inflation Adjustment:

00.030.1

Run Simulation

